C6.6 Other requirements

This section describes other requirements that apply to the interconnect. It contains the following sections:

- *Non-blocking requirements*
- *Permitted transaction modifications* on page C6-238
- *Speculative reads* on page C6-238.

C6.6.1 Non-blocking requirements

To ensure transactions always progress through a system, the protocol defines rules for snooped masters and the interconnect. The rules stipulate which transactions must always make progress and which transactions can wait for others to complete.

The rules for snooped masters are defined in Chapter C5 *Snoop Transactions*. See *Non-blocking requirements for a snooped master* on page C5-217.

To ensure transactions always progress through a system, the following rules apply for an interconnect:

- The following transactions must progress to any address without requiring any pending snoop transactions to make progress: 不需要Snoop请求
  - WriteNoSnoop
  - WriteBack
  - WriteClean
  - Evict.

  Note

  None of these transactions require an associated snoop transaction.

- An interconnect is permitted to wait for a snoop transaction to complete before it progresses the following transactions:
  - Any transaction to any address issued on the AR channel.
  - WriteUnique or WriteLineUnique transactions to any address issued on the AW channel.

    Note

    See also *Restrictions on WriteUnique and WriteLineUnique Usage* on page C4-205.

The diagram in Figure C6-3 shows the non-blocking requirements.

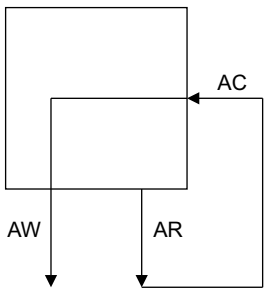


Figure C6-3 Required non-blocking transaction channel ordering

In summary, the requirements are: AC可以阻塞AR
AW可以阻塞AC

- any transaction on the AR channel can be stalled waiting for a transaction on the AC channel
- any snoop transaction on the AC channel can be stalled waiting for a write transaction on the AW channel, except for a WriteUnique or WriteLineUnique transaction.

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C6.6.2 Permitted transaction modifications

An interconnect is permitted to modify transactions as follows:

- a transaction can be broken into multiple transactions 一个请求拆分成多个请求
- multiple transactions can be merged into a single transaction. 多个请求压缩成一个请求
- a read transaction can fetch more data than required 一个Read请求可以读取更多的数据
- a write transaction can access a larger address range than required, making use of Write strobes to ensure that only the required memory locations are updated
- in each generated transaction, the following signals can be modified:
  - the transfer address, **AxADDR**
  - the burst size, **AxSIZE**
  - the burst length, **AxLEN**
  - the burst type, **AxBURST**.

Note

- The interconnect behavior must be consistent with that of a transaction that has the Modifiable attribute, **AxCACHE[1]**, asserted. See *Modifiable transactions* on page A4-61.
- A modification to a transaction by the interconnect is not seen by any master in the system.

C6.6.3 Speculative reads 预测性读取

A master in the ACE protocol is permitted to carry out a read of a cache line that it already holds in its cache. This is referred to as a Speculative Read. ？？？

A master issuing a speculative read must ensure that:

- the transaction uses the correct shareability and cacheability attributes for the address location
- it uses its cached version of the data and not the data returned by the speculative read.

Note

It is required that a master uses its cached version because this could be in the Dirty state and therefore no other valid copies of the cache line exist.

An interconnect must consider that a master might be carrying out a speculative read, as it is not explicit in the transaction. The interconnect must ensure that it does not use data obtained by a speculative read to service another transaction.

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